Sahaj Bhadja

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Summary

MSCS candidate at USC with 2+ years developing **C++** and **Python** tooling, automated build/test workflows, and performance-critical algorithms on **Linux**, with operating systems and algorithms coursework and cross-functional delivery experience.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, OOP, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built automated **Python** pipelines processing 5K+ monthly jobs with reconciliation and validation gates, maintaining 99%+ reliability through **CI/CD** releases and documented workflows for cross-team handoffs.
- Improved latency **1.2s to 350ms** for 500+ users via algorithmic **PostgreSQL** query and indexing optimization, structured logging, and monitoring hooks accelerating production defect triage.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed production **Python** services with concurrent batch execution on **AWS**, handling 10K+ daily requests with **GitHub Actions CI/CD** validation and stable throughput during peak processing workloads.
- Collaborated in code reviews with cross-functional partners, reducing feedback turnaround **60%** while independently debugging production failures and improving tool maintainability and documentation.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform tooling with **Jenkins CI/CD**, nightly VM automation, and 20+ **PyTest** suites across **Linux**, Windows, and macOS, boosting pipeline efficiency **68%** with Microsoft/GitHub partners.
- Optimized integration components using OOP and algorithmic payload reduction, improving response times **30%** while documenting automated flows to reduce manual validation and accelerate cross-team deliverables.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Automated build, test, and deployment with **Jenkins CI/CD** and **Docker** for eight microservices, cutting release time **50%** while improving flow reliability through error tracking and structured logging.
- Built data sync for 2M+ records on **Azure SQL** with schema design and query tuning, strengthening performance for high-volume distributed processing pipelines across production environments.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **Flask API** tooling for 50K+ records with sub-200ms responses across 12+ endpoints, documenting system behavior and collaborating on integration requirements for cross-platform delivery workflows.
- Reduced manual handling **70%** by diagnosing production defects through logging and monitoring while independently unblocking integration issues and supporting automated webhook-driven processing.

Skills

Languages: C++, Python, Java, Bash, SQL, JavaScript
Algorithms & Systems: data structures, OOP, performance optimization, operating systems, Linux
Automation: Jenkins, GitHub Actions, CI/CD, Docker, Git, Bash scripting, validation pipelines
Testing & Quality: PyTest, automated testing, debugging, logging, production triage, documentation
Practices: Agile, cross-functional collaboration, code reviews, ETL pipelines, microservices, AWS, Azure

Projects

CPU-scheduling Simulator Live | *C++, Algorithms, Operating Systems*
- Implemented FCFS, SJF, SRT, RR, priority, and MLQ scheduling in **C++** with edge-case handling across 1K+ process combinations, reducing average turnaround time **25%** in benchmark scenarios.

Linguistix Code | *Python, AWS ECS, Docker, CI/CD*
- Built containerized processing tooling on **AWS ECS/Fargate** with **Docker** and CodePipeline **CI/CD**, automated validation gates, and production troubleshooting across deployment pipelines.

Chat2SQL Code | *Java, TypeScript, Docker, Cross-Platform*
- Developed schema-aware SQL transformation tooling with validation across 10+ database systems on **Java** and **TypeScript**, supporting **Docker** deployment on Windows, macOS, and **Linux**.